Practice workbook · Decomposition and abstraction

Computational thinking for problem analysis: recognising, defining or explaining decomposition and abstraction in a scenario (planning comments that split a task, a library routine that hides detail).

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-13

1. Question 4 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

4 Eli is designing a three-throw dice game to play on a computer.

He uses abstraction and decomposition to help him.

(a) **Figure 5** shows how Eli uses comments to plan code before writing it.

Eli's preferred language uses a double-slash // to write comments.

```
1  // roll dice 3 times
2
3     // display number rolled
4
5     // update subtotal
6
7     // check for roll of 3 or 6 and increment booster
8
9  // calculate and output final score
10
```

Figure 5

State why this is a type of decomposition.

(1)

(b) Eli decides to use a library subprogram to generate random numbers that model a dice throw.

Explain why this is an example of abstraction.

(2)

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(c) Eli's program has these requirements.

- A roll of the dice is a random number between 1 and 6.
- The dice is rolled three times.
- The numbers rolled are added to a subtotal.
- The final score is calculated by multiplying the subtotal by a booster value.
- The booster value starts at 1.
- Each time a 3 or 6 is rolled, the booster value is incremented by 1.
- The game outputs the value of each roll, and the final score.

Figure 6 shows a trace table for one execution of the program.

Output	Subtotal	Booster
Rolled: 3	0 + 3 = 3	3 rolled, so incremented to 2
Rolled: 2	3 + 2 = 5	Neither 3 nor 6 rolled
Rolled: 6	5 + 6 = 11	6 rolled, so incremented to 3
Score 33	Subtotal (11) * Booster (3) = 33	

Figure 6

Open **Q04c** in the code editor.

Write the code to create this program.

You must use the layout, variables and constants given in **Q04c**.

Do not add any further functionality.

Save your code as **Q04cFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 4 = 11 marks)

1. Question 4 · part b · 2 marks · Working and answer

Working and answer

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2. Question 1 · part a · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part a only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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P 7 5 7 3 7 A 0 2 1 6

Working and answer

3. Question 4 · part a · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a only.

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Figure 5

State why this is a type of decomposition.

(1)

(b) Eli decides to use a library subprogram to generate random numbers that model a dice throw.

Explain why this is an example of abstraction.

(2)

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10

P 7 8 6 3 9 A 0 1 0 2 0

3. Question 4 · part a · 1 mark · Working and answer

Working and answer

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